

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Third Semester of B. Tech. (CL) Examination
November-2013**

CL204.01 Engineering Geology

Date: 28.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Sections I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a)** Write a short note on any two of the following. [04]
i. Seismic Zonation ii. Branches of Geology iii. Physical weathering
- (b)** Draw a neat sketch of interior of earth and label the major divisions and [03]
disconformities.
- Q - 2 (a)** Explain the depositional features produced by the action of river. [05]
(b) Explain in detail the process associated with denudation of rock. [04]
(c) Briefly explain the engineering difficulties in a Karst topography area. [05]

OR

- Q - 2 (a)** Explain drainage patterns and the importance of studying drainage pattern in civil [06]
engineering.
- (b)** Describe the different methods by which we can get the information of internal [04]
structure of earth.
- (c)** Explain earthquake waves and methods of locating earthquakes. [04]
- Q - 3 (a)** With help of neat diagram explain different forms of igneous rocks. [05]
(b) Explain the use of Mohs' Scale of Hardness. [05]
(c) Name any three ore minerals and briefly explain its physical properties. [04]

OR

- Q - 3 (a)** Explain briefly colour, luster and streak properties of minerals. [05]
(b) Explain gneissosity and schistosity in metamorphic rocks. [04]
(c) Explain the structures seen in Sedimentary rocks. [05]

SECTION – II

Q - 4(a) Distinguish between the following pairs with neat sketch. [05]
(i) Normal fault and Reverse fault (ii) Anticline and Syncline fold.

(b) State differences between joints and fault. Write note on occurrence of joints. [05]

(c) Define fold and folding. Describe causes of folding. [04]

OR

Q - 4 (a) 'Type of a fault developed in nature is a function of the three principal stress directions (major, intermediate and minor stress orientation)'. Justify the statement and draw sketches. [05]

(b) Write note on (i) Causes of joints (ii) Evidences of unconformity [05]

(c) State differences between map and plan. Define contour maps, geological maps and topographic maps. [04]

Q -5 (a) What does mass movement mean? State the internal factors affecting the mass movement. Suggest remedial measures to control mass movement. [07]

(b) Write notes on following geological characters that need to be investigated for dam structure. (draw neat sketch if necessary) [07]

(i) The depth of the bed rock (ii) The nature of the bed rock (iii) The structural disposition of rock.

OR

Q -5 (a) What are the problems encountered and measures adopted during tunneling in soft formations as well as hard rock? [05]

(b) Explain dam and types of dams. Also explain under what geological considerations an arch dam is preferred? [05]

(c) Enlist and explain the various information obtained from geological maps useful for the site selection in different civil structures. [04]

Q-6 (a) Explain the geological importance for the site selection of highways. [05]

(b) Write objectives of engineering geology. [02]
